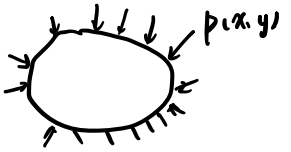
有限元自学教程第四讲

弹性力学力方程



① 平衡 $\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} = 0$, $\frac{\partial \sigma_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} = 0$

② 几何 $\varepsilon_{xx} = \frac{\partial u}{\partial x}$, $\varepsilon_{yy} = \frac{\partial v}{\partial y}$, $\gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}$

③ 胡克定律 $\varepsilon_{xx} = \frac{1}{E}(\sigma_{xx} - \nu\sigma_{yy})$ $\varepsilon_{yy} = \frac{1}{E}(\sigma_{yy} - \nu\sigma_{xx})$

$$\gamma_{xy} = \frac{1}{G}\sigma_{xy} = \frac{2(1+\nu)}{E}\sigma_{xy}$$

$$\begin{pmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{xy} \end{pmatrix} = \begin{pmatrix} \frac{1}{E} & -\frac{\nu}{E} & 0 \\ -\frac{\nu}{E} & \frac{1}{E} & 0 \\ 0 & 0 & \frac{2(1+\nu)}{E} \end{pmatrix} \begin{pmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{xy} \end{pmatrix}$$

$\vec{\varepsilon} = D\vec{\sigma}$ $\vec{\varepsilon} = (\varepsilon_{xx}, \varepsilon_{yy}, \varepsilon_{xy})$ $\sigma = (\sigma_{xx}, \sigma_{yy}, \sigma_{xy})$

④ 边界条件 $\sigma_{xx}\cos\alpha + \sigma_{xy}\cos\beta = p_x$ $\sigma_{xy}\cos\alpha + \sigma_{yy}\cos\beta = p_y$ (在边界上)

$u = 0$, $v = 0$ (在约束边界上)

补充：弹性力学方程要求 $E > 0$, $-1 < \nu < 0.5$

解存在，解唯一，$\Rightarrow$ $E>0$, $-1<\nu<0.5$，证明略

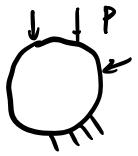
力 $\Rightarrow$ 能量 能量守恒

弹性势能 $\Rightarrow$ 应变能

$\|$ $\|$

$\frac{1}{2}kx^2$ $\iiint u_0\,dxdydz$ 或 $\iint u_0\,dxdy$（平面）

u_0 为应变能密度（单位体积的应变能）



设位移为 u、v（u、v 为 x、y 的函数）

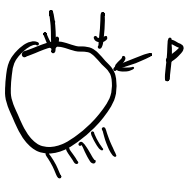
外力功：$W=\iint p_x ds\,u+p_y ds\,v=\iint(p_x u+p_y v)\,dS$

能量守恒 $W=E_p$ $E_p=\iint u_0\,dxdy$ 能量方程？ ✗

$W=E_k+E_p$



会振荡，有动能存在



$W=E_p+E_k$

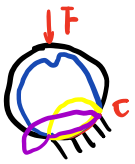
稳定状态 ⇒ E_k 因为摩擦，阻尼等因素衰减为 0 时的状态

$W\neq E_p$ E_k 不好算

需要新的物理规律：最小势能原理

平衡 ⇒ 稳定 ⇒ 能量最低 ⇒ 最小势能原理

定义： 真实位移：弹性体稳定时的位移



可能的位移：任意满足位移边界条件的位移

真实位移 ∈ 可能的位移

最小势能原理：在一切可能的位移中，真实位移总使系统的势能取极小值

$$E_p = E_{p外} + E_{p内}$$

$E_{p内}$ 为应变能，$E_{p外} = -W_{外}$

$$E_p = E_{p内} - W_{外} = \iiint u_0 dx dy dz - \iint (p_x u + p_y v + p_z w) dS$$

$$= \iint u_0 dx dy - \iint (p_x u + p_y v) dS$$



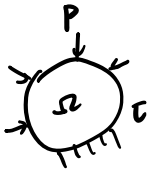
力 ⇒ 牛顿运动定律　　应力 ⇒ 应变 ⇒ 弹性力学力方程

功 ⇒ 能 ⇒ 能量守恒　　虚位移 ⇒ 虚功原理 ⇒ 最小势能原理

$\vec{F} \cdot \vec{s}$　　泛函与变分法

$$E_p = \iiint_\Omega \frac{1}{2}(\sigma_{xx}\varepsilon_{xx} + \sigma_{yy}\varepsilon_{yy} + \sigma_{zz}\varepsilon_{zz} + \sigma_{xy}\gamma_{xy} + \sigma_{yz}\gamma_{yz} + \sigma_{zx}\gamma_{zx}) dV$$

$$- \iint_\Gamma (p_x u + p_y v + p_z w) dS$$



二维：$\iint_\Omega \frac{1}{2}(\sigma_{xx}\varepsilon_{xx} + \sigma_{yy}\varepsilon_{yy} + \sigma_{xy}\gamma_{xy}) dV - \iint_\Gamma (p_x u + p_y v) dS$

证明最小势能原理

① 证明 $u_0 = \frac{1}{2}(\sigma_{xx}\varepsilon_{xx} + \sigma_{yy}\varepsilon_{yy} + \sigma_{xy}\gamma_{xy})$

② 证明：若 $u, v, \sigma_{xx}, \sigma_{yy}, \sigma_{xy}, \varepsilon_{xx}, \varepsilon_{yy}, \varepsilon_{xy}$ 是真实位移，则 E_p 取极小值

$W_{外} = E_k + E_p \qquad dW_{外} = dE_k + dE_p$

$W_{外} = \iint_\Gamma (p_x u + p_y v)\,dS \qquad dW_{外} = \iint_\Gamma (p_x\,du + p_y\,dv)\,dS$

$E_k = \frac{1}{2}mv^2 = \frac{1}{2}\iiint dm \cdot v^2 = \frac{1}{2}\iiint \rho\,dV\cdot(\dot{u}^2+\dot{v}^2) = \iiint \frac{1}{2}\rho(\dot{u}^2+\dot{v}^2)\,dV$

$dE_k = \iiint \frac{1}{2}\rho[d(\dot{u}^2) + d(\dot{v}^2)]\,dV$

$E_p = \iiint u_0\,dV \qquad dE_p = \iiint (du_0)\,dV$

$dW_{外} = dE_k + dE_p$

$dW_{外} = \iint_\Gamma (p_x\,du + p_y\,dv)\,dS$ 边界条件 $\sigma_x\cos\alpha + \tau_{xy}\cos\beta = p_x$，$\tau_{xy}\cos\alpha + \sigma_y\cos\beta = p_y$

$dW_{外} = \iint_\Gamma [(\sigma_x\,du + \tau_{xy}\,dv)\cos\alpha + (\tau_{xy}\,du + \sigma_y\,dv)\cos\beta]\,dS$

复习：高斯公式（面积分⇔体积分）

$$\iiint \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}\right)dV = \iint (P\cos\alpha + Q\cos\beta + R\cos\gamma)\,dS$$



$(\cos\alpha, \cos\beta, \cos\gamma)$ 为 Ω 表面 Γ 的法向量的方向余弦

令 $R=0 \Rightarrow \iiint \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}\right)dV = \iint (P\cos\alpha + Q\cos\beta)\,dS$

逆用高斯公式

$dW_{外} = \iiint_\Omega \left[\frac{\partial}{\partial x}(\sigma_x\,du + \tau_{xy}\,dv) + \frac{\partial}{\partial y}(\tau_{xy}\,du + \sigma_y\,dv)\right]dV$

展开 $dW_{外} = \iiint_\Omega \frac{\partial \sigma_x}{\partial x}\,du + \sigma_x\,d(\varepsilon_x) + \frac{\partial \tau_{xy}}{\partial x}\,dv + \tau_{xy}\,d\left(\frac{\partial v}{\partial y}\right) + \cdots$

$\frac{\partial du}{\partial x} = d\left(\frac{\partial u}{\partial x}\right) = d(\varepsilon_x)$ 几何方程，及微分与偏导交换

$$dW_{外} = \iiint_{\Omega} \left[\left(\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} \right) du + \left(\frac{\partial \sigma_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} \right) dv + \sigma_{xx} d\varepsilon_{xx} + \sigma_{xy} d\gamma_{xy} + \sigma_{yy} d\varepsilon_{yy} \right] dV$$

平衡方程

$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} = 0$ (非平衡, 有 E_k) $\frac{\partial \sigma_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} = \rho \ddot{u}$

$$dW_{外} = \iiint_{\Omega} \left[d\left(\frac{1}{2} \rho (\dot{u}^2 + \dot{v}^2) \right) + \sigma_{xx} d\varepsilon_{xx} + \sigma_{xy} d\gamma_{xy} + \sigma_{yy} d\varepsilon_{yy} \right] dV$$

$$dW_{外} = dE_k + \iiint (\sigma_{xx} d\varepsilon_{xx} + \sigma_{yy} d\varepsilon_{yy} \quad \sigma_{xy} d\gamma_{xy}) dV \qquad dE_p \qquad dE_p = \iiint (du_0) dV$$

$$\therefore du_0 = \sigma_{xx} d\varepsilon_{xx} + \sigma_{yy} d\varepsilon_{yy} + \sigma_{xy} d\gamma_{xy} = (\sigma_{xx} \quad \sigma_{yy} \quad \sigma_{xy}) \begin{pmatrix} d\varepsilon_{xx} \\ d\varepsilon_{yy} \\ d\gamma_{xy} \end{pmatrix}$$

$\vec{\sigma} = (\sigma_{xx} \quad \sigma_{yy} \quad \sigma_{xy})$ $\vec{\varepsilon} = (\varepsilon_{xx} \quad \varepsilon_{yy} \quad \gamma_{xy})$

$\therefore du_0 = \vec{\sigma} (d\vec{\varepsilon})^T$ 又 $\vec{\varepsilon} = D\vec{\sigma}$ (胡克定律) $\therefore \vec{\sigma} = D^{-1}\vec{\varepsilon}$

$$\therefore du_0 = D^{-1}\vec{\varepsilon}(d\vec{\varepsilon})^T = \frac{1}{2} d\left(D^{-1}\vec{\varepsilon}(\vec{\varepsilon})^T \right) = d\left(\frac{1}{2} \vec{\sigma} \vec{\varepsilon}^T \right)$$

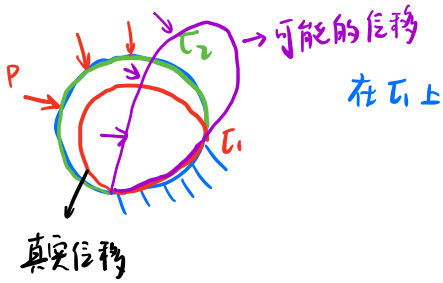
$$\therefore u_0 = \frac{1}{2} \vec{\sigma} \vec{\varepsilon}^T$$

$$u_0 = \frac{1}{2} (\sigma_{xx} \varepsilon_{xx} + \sigma_{yy} \varepsilon_{yy} + \sigma_{xy} \gamma_{xy})$$

② $E_p = \iiint u_0\,dV - \iint (p_x u + p_y v)\,dS = \iiint \frac{1}{2}(\sigma_x \varepsilon_x + \sigma_y \varepsilon_y + \tau_{xy}\gamma_{xy})\,dV - \iint (p_x u + p_y v)\,dS$

设 u, v 为真实位移，u', v' 为可能的位移

设 $u' = u + u^*$，$v' = v + v^*$　　u^*, v^* 为满足位移边界条件的任意函数



在 Γ_1 上，$u^* = v^* = 0$

u^*, v^* 可以不满足应力边界条件，任意函数

u, v 对应的系统势能为 $E_p = \iiint u_0\,dV - \iint (p_x u + p_y v)\,dS$

u', v' 对应的系统势能为 $E_p' = \iiint u_0'\,dV - \iint (p_x u' + p_y v')\,dS$

最小势能原理 $\Rightarrow$ $E_p' \geq E_p$（仅当 $u^* = v^* = 0$ 时 $E_p' = E_p$）

暴力证明：$(AB)C = A(BC)$ $\begin{pmatrix} a_1 & a_2 & a_3 \\ \vdots & & \end{pmatrix}$ $\begin{pmatrix} b_1 & b_2 & b_3 \\ & & \end{pmatrix}$ $\begin{pmatrix} c_1 & c_2 & c_3 \\ & & \end{pmatrix}$

技巧性很低，计算量极大

(二维证明)

证明 $E_p' > E_p$

$$E_p' = \iiint \frac{1}{2}[(\sigma_{xx}+\sigma_{xx}^*)(\varepsilon_{xx}+\varepsilon_{xx}^*)+(\sigma_{yy}+\sigma_{yy}^*)(\varepsilon_{yy}+\varepsilon_{yy}^*)+(\sigma_{xy}+\sigma_{xy}^*)(\varepsilon_{xy}+\varepsilon_{xy}^*)]\,dV$$

$$-\iint[P_x(u+u^*)+P_y(v+v^*)]\,dS$$

展开：16次乘法，16项

$$E_p' = \iiint \frac{1}{2}[\sigma_{xx}\varepsilon_{xx}+\sigma_{xx}\varepsilon_{xx}^*+\sigma_{xx}^*\varepsilon_{xx}+\sigma_{xx}^*\varepsilon_{xx}^*+\sigma_{yy}\varepsilon_{yy}+\sigma_{yy}^*\varepsilon_{yy}+\sigma_{yy}\varepsilon_{yy}^*+\sigma_{yy}^*\varepsilon_{yy}^*$$

$$+\sigma_{xy}\gamma_{xy}+\sigma_{xy}\gamma_{xy}^*+\sigma_{xy}^*\gamma_{xy}+\sigma_{xy}^*\gamma_{xy}^*]\,dV-\iint(P_xu+P_xu^*+P_yv+P_yv^*)\,dS$$

$$=\underbrace{\iiint \frac{1}{2}[\sigma_{xx}\varepsilon_{xx}+\sigma_{yy}\varepsilon_{yy}+\sigma_{xy}\gamma_{xy}]\,dV-\iint(P_xu+P_yv)\,dS}_{E_p}\quad(\text{五项})$$

$$+\underbrace{\iiint \frac{1}{2}[\sigma_{xx}^*\varepsilon_{xx}^*+\sigma_{yy}^*\varepsilon_{yy}^*+\sigma_{xy}^*\gamma_{xy}^*]\,dV}_{>0}\quad(\text{三项})$$

$$+\underbrace{\iiint \frac{1}{2}[\sigma_{xx}\varepsilon_{xx}^*+\sigma_{xx}^*\varepsilon_{xx}+\sigma_{yy}\varepsilon_{yy}^*+\sigma_{yy}^*\varepsilon_{yy}+\sigma_{xy}\gamma_{xy}^*+\sigma_{xy}^*\gamma_{xy}]\,dV-\iint(P_xu^*+P_yv^*)\,dS}_{=0}$$

①②③④⑤

$E_p' = E_p + \text{正} + 0$　最小势能原理得证　(八项)

1) $\iiint \frac{1}{2}[\sigma_{xx}^*\varepsilon_{xx}^*+\sigma_{yy}^*\varepsilon_{yy}^*+\sigma_{xy}^*\gamma_{xy}^*]\,dV$

$\varepsilon_{xx}^* = \frac{1}{E}(\sigma_{xx}^*-\nu\sigma_{yy}^*)$　$\varepsilon_{yy}^* = \frac{1}{E}(\sigma_{yy}^*-\nu\sigma_{xx}^*)$　$\gamma_{xy}^* = \frac{2(1+\nu)}{E}\sigma_{xy}^*$

$$\frac{1}{2E}\iiint(\sigma_{xx}^*(\sigma_{xx}^*-\nu\sigma_{yy}^*)+\sigma_{yy}^*(\sigma_{yy}^*-\nu\sigma_{xx}^*)+2(1+\nu)(\sigma_{xy}^*)^2)\,dV$$

$$=\frac{1}{2E}\iiint[(\sigma_{xx}^*)^2+(\sigma_{yy}^*)^2-2\nu\sigma_{xx}^*\sigma_{yy}^*+2(1+\nu)(\sigma_{xy}^*)^2]\,dV$$

$$=\frac{1}{2E}\iiint[\underbrace{(\sigma_{xx}^*-\nu\sigma_{yy}^*)^2}_{>0}+\underbrace{(1-\nu^2)}_{>0}\underbrace{(\sigma_{yy}^*)^2}_{>0}+\underbrace{2(1+\nu)}_{>0}\underbrace{(\sigma_{xy}^*)^2}_{>0}]\,dV \quad \therefore >0$$

$-2\nu<0$

仅当 $\sigma_{xx}-\sigma_{yy}=\sigma_{xy}=0$ 时 $\iiint=0$ $u^*=0,\ v^*=0$

2) 证明 $\iiint \frac{1}{2}(\sigma_{xx}\varepsilon_{xx}^* + \sigma_{xx}^*\varepsilon_{xx} + \sigma_{yy}\varepsilon_{yy}^* + \sigma_{yy}^*\varepsilon_{yy} + \sigma_{xy}^*\varepsilon_{xy} + \sigma_{xy}\varepsilon_{xy}^*)dV - \iint(P_x u^* + P_y v^*)dS = 0$

$\sigma_{xx}\varepsilon_{xx}^* = \sigma_{xx}\left(\frac{1}{E}(\sigma_{xx}^* - v\sigma_{yy}^*)\right) = \frac{1}{E}(\sigma_{xx}\sigma_{xx}^* - v\sigma_{xx}\sigma_{yy}^*)$

$\sigma_{xx}^*\varepsilon_{xx} = \sigma_{xx}^*\left(\frac{1}{E}(\sigma_{xx} - v\sigma_{yy})\right) = \frac{1}{E}(\sigma_{xx}^*\sigma_{xx} - v\sigma_{xx}^*\sigma_{yy})$ $\sigma_{xx}^*\varepsilon_{xx} \neq \sigma_{xx}\varepsilon_{xx}^*$

同理 $\sigma_{yy}^*\varepsilon_{yy} = \frac{1}{E}(\sigma_{yy}^*\sigma_{yy} - v\sigma_{yy}^*\sigma_{xx})$

$\sigma_{yy}\varepsilon_{yy}^* = \frac{1}{E}(\sigma_{yy}\sigma_{yy}^* - v\sigma_{yy}\sigma_{xx}^*)$ $\sigma_{yy}^*\varepsilon_{yy} \neq \sigma_{yy}\varepsilon_{yy}^*$

但是 $\sigma_{xx}\varepsilon_{xx}^* + \sigma_{xx}^*\varepsilon_{xx} + \sigma_{yy}^*\varepsilon_{yy} + \sigma_{yy}\varepsilon_{yy}^* = \frac{1}{E}[2\sigma_{xx}\sigma_{xx}^* + 2\sigma_{yy}\sigma_{yy}^* - 2v(\sigma_{yy}^*\sigma_{xx} + \sigma_{yy}\sigma_{xx}^*)]$

$= \frac{1}{E}(2\sigma_{xx}\sigma_{xx}^* - 2v\sigma_{xx}\sigma_{yy}^* + 2\sigma_{yy}\sigma_{yy}^* - 2v\sigma_{yy}\sigma_{xx}^*)$

$= 2\sigma_{xx}\frac{1}{E}(\sigma_{xx}^* - v\sigma_{yy}^*) + 2\sigma_{yy}\frac{1}{E}(\sigma_{yy}^* - v\sigma_{xx}^*) = 2\sigma_{xx}\varepsilon_{xx}^* + 2\sigma_{yy}\varepsilon_{yy}^*$

胡克定律

且 $\sigma_{xy}\gamma_{xy}^* = \sigma_{xy}^*\gamma_{xy}$ $\left(\gamma_{xy} = \frac{\sigma_{xy}}{G},\ \gamma_{xy}^* = \frac{\sigma_{xy}^*}{G},\ G = \frac{E}{2(1+v)}\right)$

① 胡克定律

$\therefore I = \iiint \frac{1}{2}(2(\sigma_{xx}\varepsilon_{xx}^* + 2\sigma_{yy}\varepsilon_{yy}^* + 2\sigma_{xy}\gamma_{xy}^*)dV - \iint(P_x u^* + P_y v^*)dS$

⇓ 几何方程 ②

$= \iiint(\sigma_{xx}\frac{\partial u^*}{\partial x} + \sigma_{yy}\frac{\partial v^*}{\partial y} + \sigma_{xy}(\frac{\partial u^*}{\partial y} + \frac{\partial v^*}{\partial x}))dV - \iint(P_x u^* + P_y v^*)dS$

$= \iiint(\frac{\partial(\sigma_{xx}u^*)}{\partial x} - u^*\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial(\sigma_{yy}v^*)}{\partial y} - v^*\cdot\frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial(\sigma_{xy}u^*)}{\partial y} - u^*\cdot\frac{\partial \sigma_{xy}}{\partial y} + \frac{\partial(\sigma_{xy}v^*)}{\partial x} - v^*\frac{\partial \sigma_{xy}}{\partial x})dV$

$- \iint(P_x u^* + P_y v^*)dS$ (10项)

$= \iiint[\frac{\partial(\sigma_{xx}u^* + \sigma_{xy}v^*)}{\partial x} + \frac{\partial(\sigma_{yy}v^* + \sigma_{xy}u^*)}{\partial y}]dV - \iint(P_x u^* + P_y v^*)dS$ (6项)

⇓ dS

$+ \iiint[u^*(\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y}) - v^*(\frac{\partial \sigma_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y})]dV$

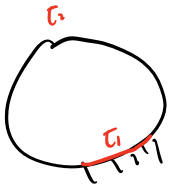
牛顿运动定律（平衡方程）③

高斯公式 $\iiint \left(\frac{\partial P}{\partial x}+\frac{\partial Q}{\partial y}\right) dV=\iint (P\cos\alpha+Q\cos\beta) dS \quad (R=0)$

其中 $\cos\alpha$, $\cos\beta$, $\cos\gamma$ 为边界的外法线的方向余弦

$$I=\iint \left[(\sigma_x u^*+\tau_{xy}v^*)\cos\alpha+(\tau_{xy}u^*+\sigma_y v^*)\cos\beta-p_x u^*-p_y v^*\right] dS$$

$$=\iint_{\Gamma_2+\Gamma_1} (\sigma_x\cos\alpha+\tau_{xy}\cos\beta-p_x)u^*+(\tau_{xy}\cos\alpha+\sigma_y\cos\beta-p_y)v^* \, dS$$



$(p_x, p_y)\in\Gamma_2$

$u^*, v^*=0$ (Γ_1 上) (位移边界条件)

$\therefore I=\iint_{\Gamma_2}(\sigma_x\cos\alpha+\tau_{xy}\cos\beta-p_x)u^*+(\tau_{xy}\cos\alpha+\sigma_y\cos\beta-p_y)v^* ds$

④ 应力边界条件

$+\iint_{\Gamma_1}(\quad) ds$

⑤ 位移边界条件

$\therefore I=0$ 最小势能原理得证

最小势能原理与弹性力学方程 ① ② ③ ④ ⑤ 完全等价 !!

三维情况的证明与二维情况思路一致 (计算量巨大)

求解 ① ② ③ ④ ⑤ ⇒ 求解 $\iiint u\,dV-\iint(p_x u+p_y v)\,dS$ 取极小值的那个解

有限元法定义：微分方程 ⇒ 泛函极值 ⇒ 近似解

最后一步

力方程 ⇒ 能量方程 ⇒ 近似解

泛函：将一个函数映射为一个数

函数：将一个数映射为一个数

弹性力学泛函 $\pi = \iiint \frac{1}{2}(\sigma_{xx}\varepsilon_{xx} + \sigma_{yy}\varepsilon_{yy} + \sigma_{xy}\gamma_{xy})\,dV - \iint (p_x u + p_y v)\,dS$